ST443 Week 2

Implementing linear regression using statsmodels

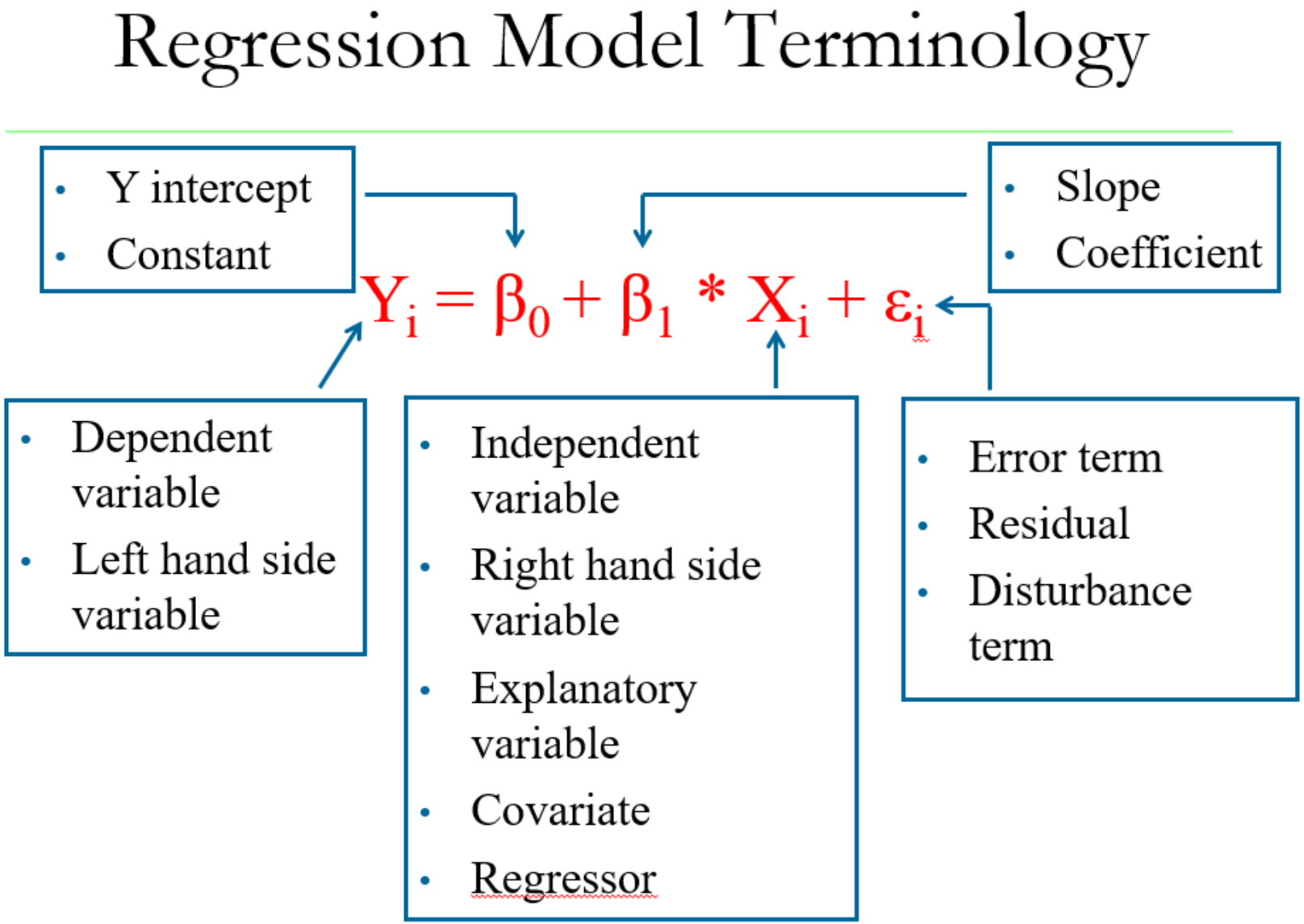
(more econometrics than ML)

6 October 2026



Notebook |Linear regression

Model: simple linear regression



Linear regression in matrix form

To develop theory for the linear regression model we need to recast it in matrix form:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

where $\mathbf{y}$ is a vector of length n, $\mathbf{X}$ is a $n \times p$ matrix containing the predictors, and $\boldsymbol{\epsilon}$ is a vector of length n that consists of the error terms. For a model that contains an intercept term:

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{22} & \cdots & x_{nk} \end{pmatrix}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{pmatrix}, \quad \boldsymbol{\epsilon} = \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{pmatrix}$$

y	x
10	2
15	7
12	4



$$\begin{aligned} 10 &= \beta_0 + 2\beta_1 + \epsilon_1, \\ 15 &= \beta_0 + 7\beta_1 + \epsilon_2, \\ 12 &= \beta_0 + 4\beta_1 + \epsilon_3. \end{aligned}$$

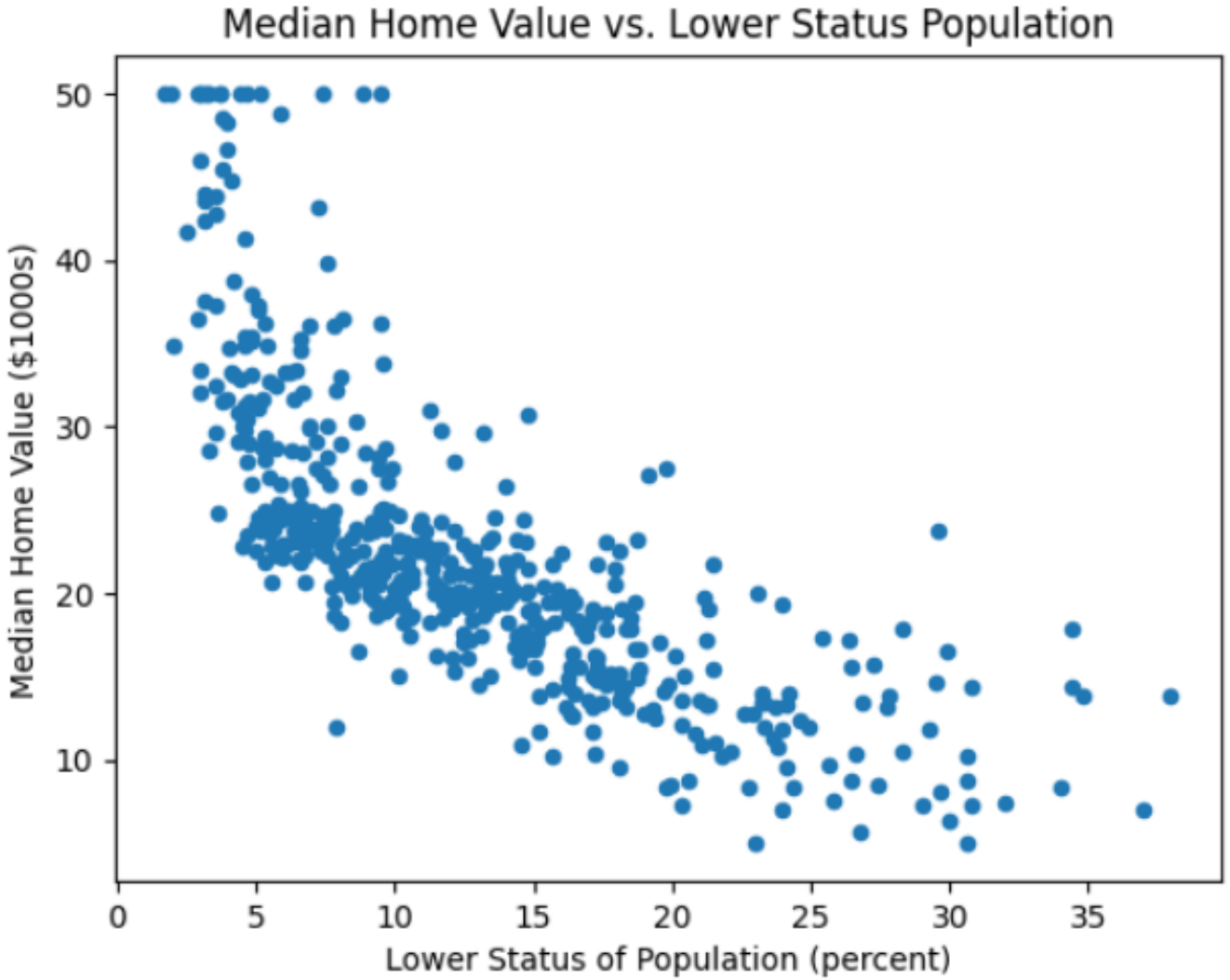
$$\mathbf{y} = \begin{pmatrix} 10 \\ 15 \\ 12 \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} 1 & 2 \\ 1 & 7 \\ 1 & 4 \end{pmatrix}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}$$

Example

The Boston Housing Dataset is a derived from information collected by the U.S. Census Service concerning housing in the area of Boston MA.

The data includes the variables:

- LSTAT - % lower status of the population
- MEDV - Median value of owner-occupied homes in $1000's


Median Home Value vs. Lower Status Population
Median Home Value ($1000s)
50
40
30
20
10
0
5
10
15
20
25
30
35
Lower Status of Population (percent)

Regress medv on lstat on the training data

	coef	std err	t	P>|t|
intercept	34.5538	0.563	61.415	0.0
lstat	-0.9500	0.039	-24.528	0.0

Fitted model

$$\hat{medv} = 34.55 - 0.95lstat$$

- Can we use this model to predict medv?
- Does the model allow us to interpret the relationship between the input and output?

statsmodel package

- Statsmodels provides statistical output for:
  - inference
  - understanding relationships

- scikit-learn is the go-to package for prediction tasks with classical ML models.
  - Classification, regression, clustering, dimension reduction

```
Boston.plot.scatter('lstat', 'medv')

# Add title and axis labels
plt.title('Median Home Value vs. Lower Status Population')
plt.xlabel('Lower Status of Population (percent)')
plt.ylabel('Median Home Value ($1000s)')

# Display the plot
plt.show()
```

```
# Create design matrix
X = pd.DataFrame({
    'intercept': np.ones(Boston.shape[0]),
    'lstat': Boston['lstat']})
```

```
X[:4]
```

	intercept	lstat
0	1.0	4.98
1	1.0	9.14
2	1.0	4.03
3	1.0	2.94

```
y = Boston['medv']

model = sm.OLS(y, X)

results = model.fit()
```

sm.OLS
Create an OLS object (but doesn't fit it yet)

Fit

- Estimates the model using OLS.
- Results: contains diagnostic output for inference (esimates, standard erros, etc..)

Advanced predictors: interactions and categorical variables

To be covered in Week 3 lecture



T/F?

A qualitative predictor may be included in a regression by creating a set of dummy variables?

A qualitative predictor may be included in a regression by creating a set of dummy variables? True

Suppose we have a factor called Status with 5 levels: *Very Low, Low, Medium, High, very high*. The design matrix X in the model:

$$y = X\beta + \epsilon$$

looks like:

	status	statusHigh	statusLow	statusMedium	statusVery High
1	Very High	0	0	0	1
2	Very High	0	0	0	1
3	High	1	0	0	0
4	High	1	0	0	0
5	Medium	0	0	1	0
6	Medium	0	0	1	0
7	Low	0	1	0	0
8	Low	0	1	0	0
9	Very Low	0	0	0	0
10	Very Low	0	0	0	0
11	Very Low	0	0	0	0

- The factor has 5 levels
- 4 dummies with a baseline is used to represent the qualitative variable
- What is the baseline here?

Categorical predictors

Categorical predictors (factors) may be included in a regression model.

A **factor** is comprised of **levels**. For example, the factor *gender* has two levels: male and female. We create a set of **dummy variable(s)** for the factor which enter the regression. The usual way to create the dummies[1] for regression is to:

- Choose a **baseline (reference) category**. Usually we set this as the group for which we wish to compare the other groups to.
- Create indicator (dummy) variables for the remaining levels.

Now we can add the set of dummies additively, or also include multiplicative terms (**interactions**).

[1] we will see another way in Week 5

In a regression of y on *gender* and another predictor, say we wish to compare the response of males to females. We make female the reference category. The regression model would be:

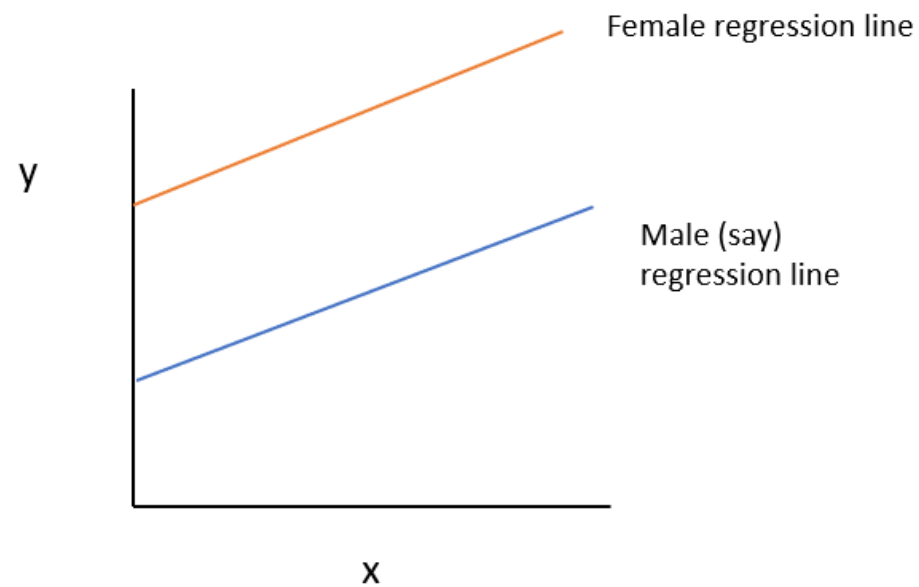
$$y_i = \beta_0 + \beta_1 D_i + \beta_2 x_i + \epsilon_i, \tag{1}$$

where $D_i = 1$ if the i-th person is male; 0 otherwise.

What is the interpretation of β_1 for a given x_i? Assuming $\mathbb{E}\epsilon_i = 0$,

$$\beta_1 = \mathbb{E}(y_i|D_i = 1) - \mathbb{E}(y_i|D_i = 0)$$

That is, β_1 is the **mean difference** in y of males **compared to** females. When the dummy variable enters in the model as in (1) it is sometimes called an **intercept dummy** due to the interpretation.

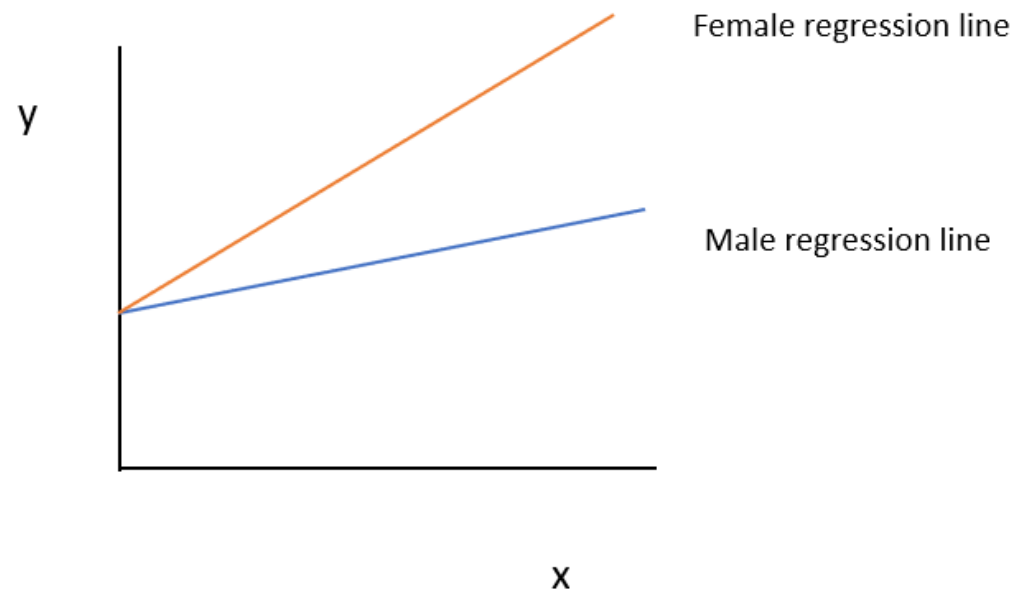


Including a **slope dummy** in a regression allows the slope parameter to be different across two groups. To illustrate, a SLR with a slope dummy D is:

$$\begin{aligned} y_i &= \beta_0 + \beta_1 x_i + \beta_3 D_i x_i + \epsilon_i \\ &= \beta_0 + (\beta_1 + \beta_3 D_i) x_i + \epsilon_i \end{aligned}$$

The term in red is an **Interaction term** between x and D.

An **interaction** between two predictors is present in a model if the effect of changing one predictor on the response is **different** for different values of the other predictor.[4]



[4]Specifically, it's a two way interaction. We can include higher order interactions.



T/F

The coefficient on an (two-way)interaction term in linear regression tells us the effect of one predictor conditional on another

Interaction term

To illustrate the interpretation of a (2-way) interaction term where the predictors in the interaction are continuous. Dropping the i subscript for clarity, consider the model:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + \epsilon$$

Say we wish to know a unit change in x_1 on $\mathbb{E}(y)$. This would be given by:

$$\frac{\partial \mathbb{E}(y)}{\partial x_1} = \beta_1 + \beta_3 x_2$$

You can see the answer depends on the value of the other variable x_2. Compare our discussion to the case where there is no interaction term.

A rule is if we include an interaction term, we should **include all lower order terms**.